psymetadata: Memory Suppression example

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```r
library(metaselection)
library(metadat)
library(psymetadata)
library(metafor)
```

```
## Loading required package: Matrix
```

```
## Loading required package: numDeriv
```

```
##
## Loading the 'metafor' package (version 4.8-0). For an
## introduction to the package please type: help(metafor)
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x tidyr::expand() masks Matrix::expand()
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## x tidyr::pack()   masks Matrix::pack()
## x tidyr::unpack() masks Matrix::unpack()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(here)
```

```
## here() starts at C:/GitHub/metaselection
```

```r
devtools::load_all()
```

```
## i Loading metaselection
```

Memory Suppression Meta-analysis

```r
dat_ex <- stramaccia2021

dat_ex <- dat_ex |>
  rename(esid = es_id, studyid = study_id) |>
  mutate(
    d = yi,
    Va = vi,
    Var_d = Va,
    sd_d = sqrt(Var_d),
    var_d = Var_d,
    p_onesided = 1 - pnorm(d / sd_d)
  )

length(unique(dat_ex$studyid)); length(unique(dat_ex$esid))
```

```
## [1] 25
```

```
## [1] 96
```

Mean Only

Unadjusted estimate: mean only

```r
ex_unadj_mean <- rma.uni(
  data = dat_ex,
  yi = d,
  sei = sd_d,
  method = "ML"
)
ex_unadj_mean
```

```
##
## Random-Effects Model (k = 96; tau^2 estimator: ML)
##
## tau^2 (estimated amount of total heterogeneity): 0.0972 (SE = 0.0216)
## tau (square root of estimated tau^2 value):      0.3117
## I^2 (total heterogeneity / total variability):   68.33%
## H^2 (total variability / sampling variability):  3.16
##
## Test for Heterogeneity:
## Q(df = 95) = 289.8412, p-val < .0001
##
## Model Results:
##
## estimate      se    zval    pval   ci.lb   ci.ub
##   0.2191  0.0397  5.5144  <.0001  0.1412  0.2970  ***
##
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

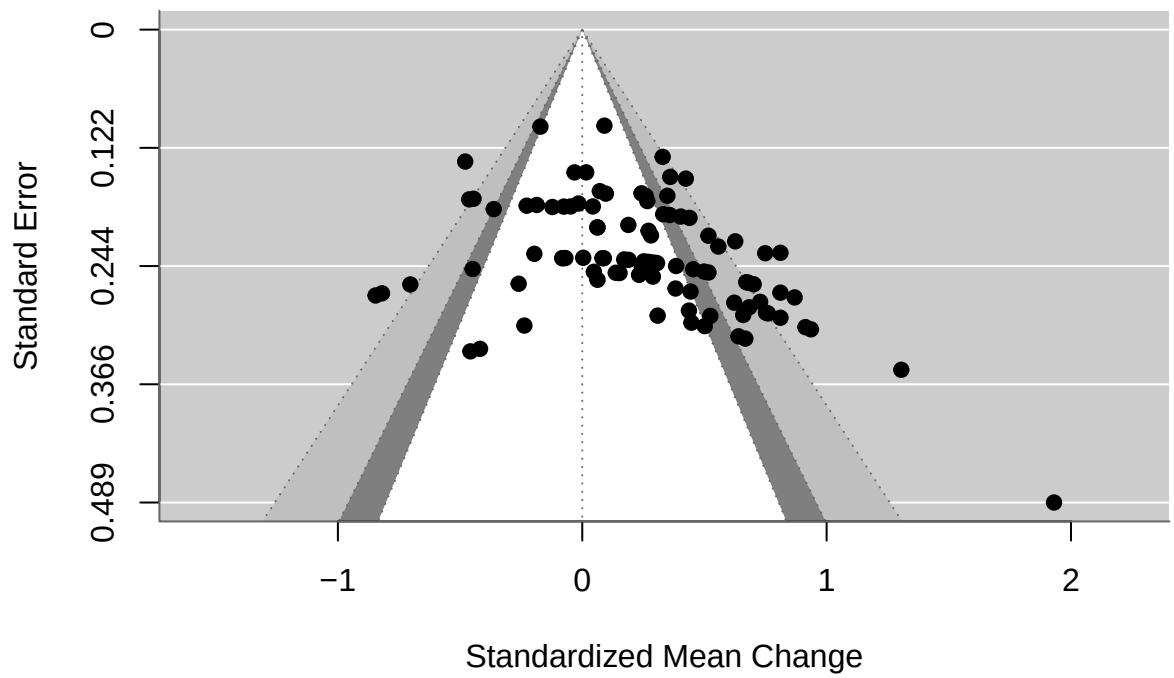
```
ex_unadj_mean_cil <- ex_unadj_mean$tau2 - 1.96*ex_unadj_mean$se.tau2
ex_unadj_mean_ciu <- ex_unadj_mean$tau2 + 1.96*ex_unadj_mean$se.tau2
ex_unadj_mean_cil; ex_unadj_mean_ciu
```

```
## [1] 0.05487575
```

```
## [1] 0.1394257
```

Forest plot

```
funnel(ex_unadj_mean, refline = 0, level = c(90, 95, 99))
```



CHE estimate: mean only

```
V_mat <- vcalc(
    vi = vi,
    cluster = studyid,
```

```r
    obs = esid,
    data = dat_ex,
    rho = 0.5,
    sparse = TRUE
  )

ex_che <- rma.mv(yi = yi,
                 V = V_mat,
                 random = ~ 1 | studyid / esid,
                 data = dat_ex,
                 sparse = TRUE,
                 digits = 3) |>
    robust(cluster = studyid, clubSandwich = TRUE)

ex_che
```

```
##
## Multivariate Meta-Analysis Model (k = 96; method: REML)
##
## Variance Components:
##
##            estim   sqrt  nlvls  fixed        factor
## sigma^2.1  0.025  0.158     25     no       studyid
## sigma^2.2  0.064  0.253     96     no  studyid/esid
##
## Test for Heterogeneity:
## Q(df = 95) = 319.112, p-val < .001
##
## Number of estimates:   96
## Number of clusters:    25
## Estimates per cluster: 2-8 (mean: 3.84, median: 4)
##
## Model Results:
##
## estimate     se¹   tval¹     df¹   pval¹  ci.lb¹  ci.ub¹
##    0.162  0.056   2.894   22.75   0.008   0.046   0.278   **
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## 1) results based on cluster-robust inference (var-cov estimator: CR2,
##    approx t-test and confidence interval, df: Satterthwaite approx)
```

Adjusted estimate in metaselection: Beta model mean only

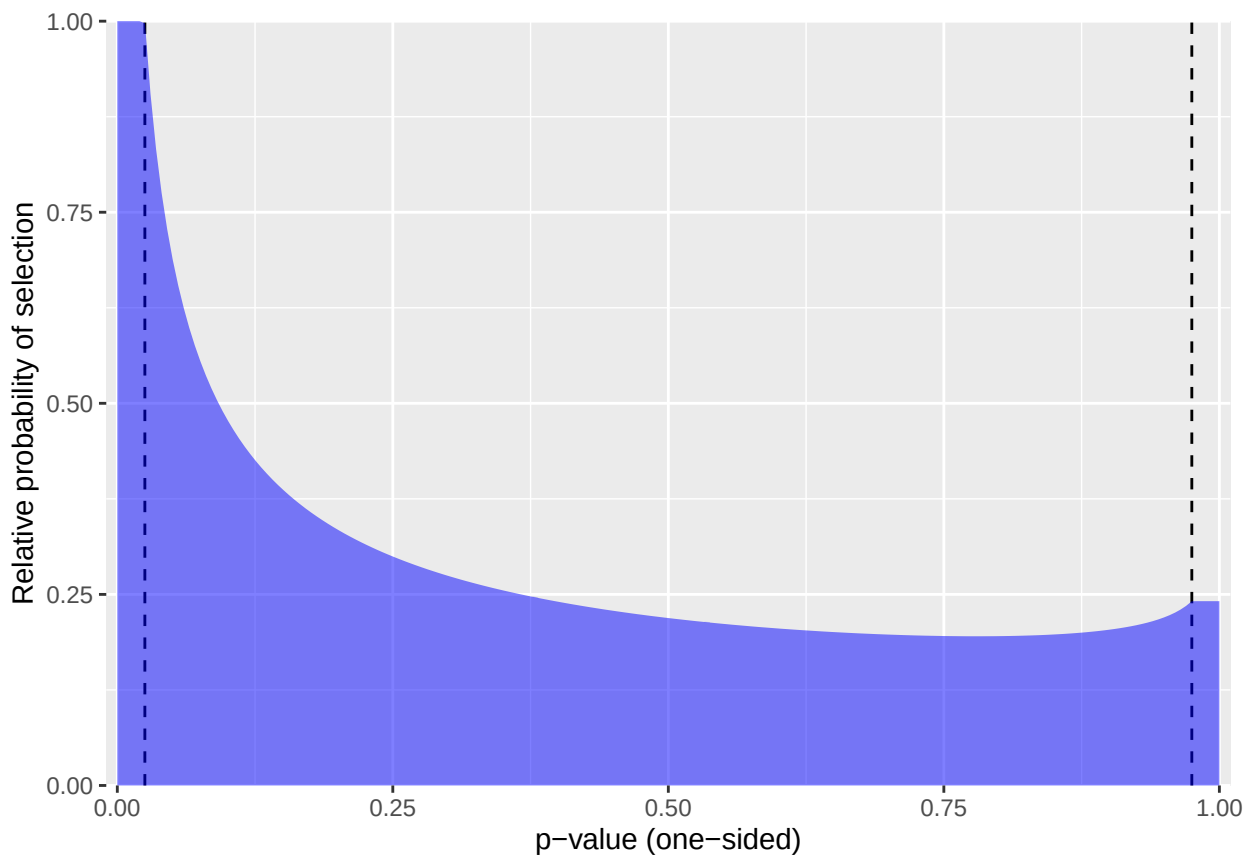
```r
ex_adj_ms_beta_mean <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  pi = p_onesided,
  cluster = studyid,
```

```
  selection_type = "beta",
  steps = c(.025,.975),
  make_sandwich = TRUE
)
ex_adj_ms_beta_mean
```

```
##    param    Est     SE p_value   CI_lo CI_hi
##     beta 0.0224 0.0941  0.8118 -0.1620 0.207
##     tau2 0.0651 0.0233      NA  0.0323 0.131
##  lambda1 0.4591 0.1818  0.0493  0.2113 0.998
##  lambda2 0.8474 0.1687  0.4054  0.5736 1.252
```

Selection plot: Beta model mean only

```
selection_plot(ex_adj_ms_beta_mean)
```



Adjusted estimate in metaselection: Step 3PSM model mean only

```
ex_adj_ms_step3_mean <- selection_model(
  dat = dat_ex,
```

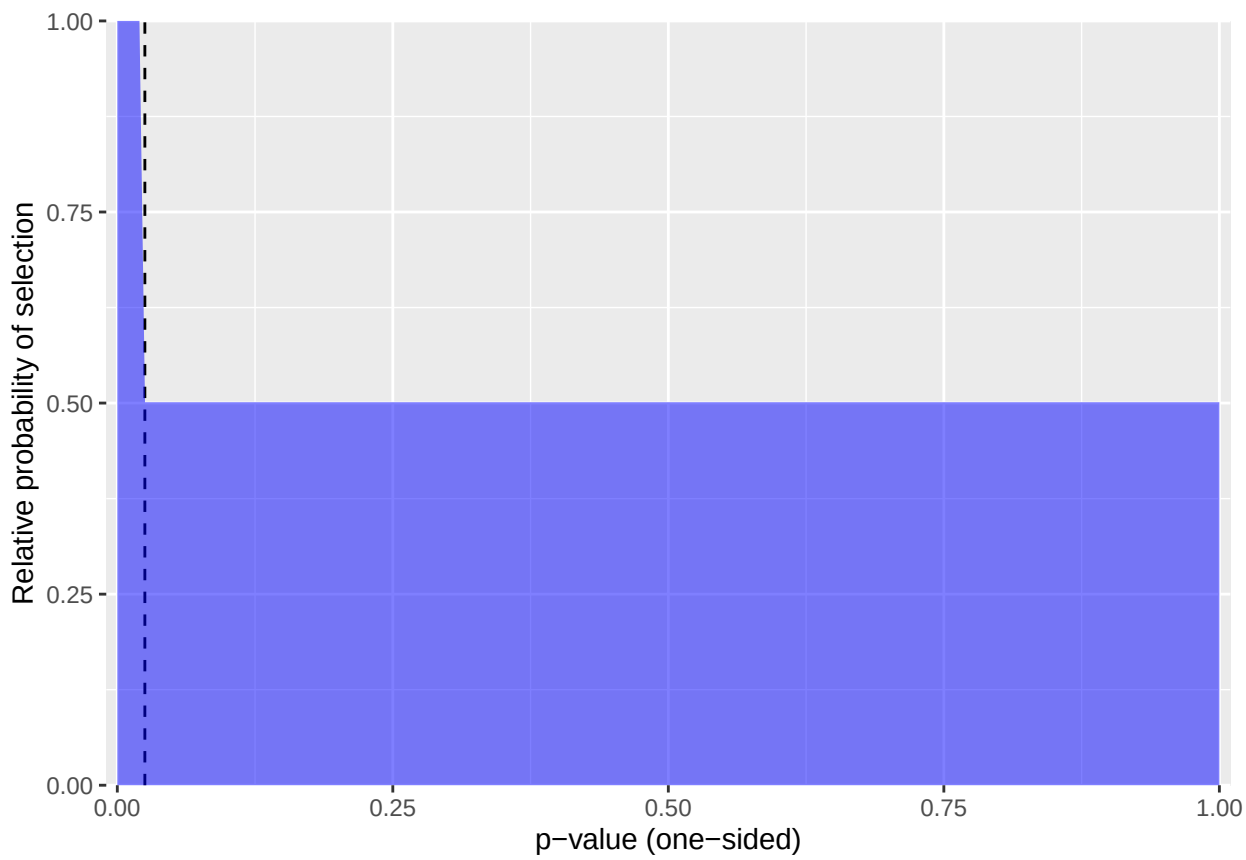
```
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = 0.025
)

ex_adj_ms_step3_mean
```

```
##    param    Est     SE p_value  CI_lo CI_hi
##     beta 0.1346 0.0562  0.0166 0.0245 0.245
##     tau2 0.0718 0.0278      NA 0.0336 0.153
##  lambda1 0.5011 0.1456  0.0174 0.2836 0.886
```

Selection plot: Step 3 PSM model mean only

```
selection_plot(ex_adj_ms_step3_mean)
```



Adjusted estimate in metaselection: Step 4PSM model mean only

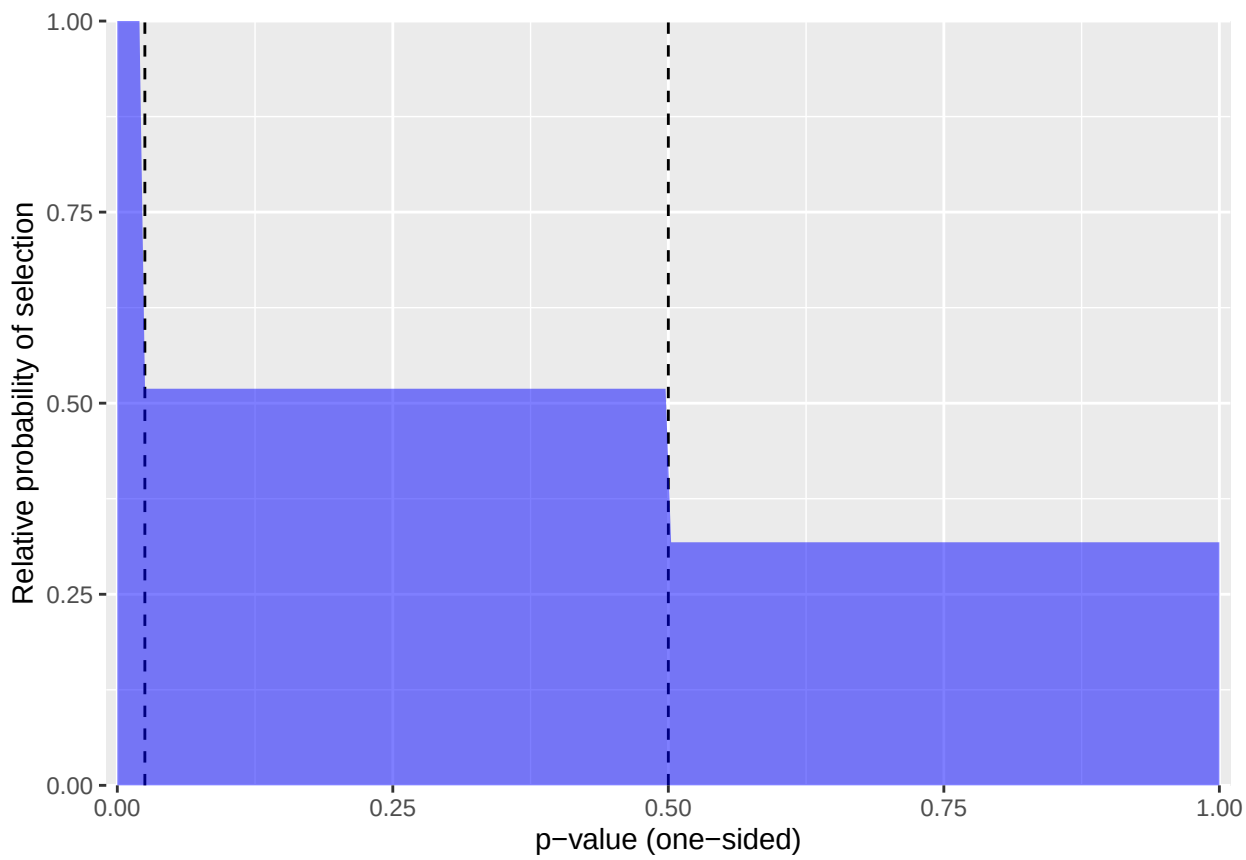
```r
ex_adj_ms_step4_mean <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = c(0.025, .5)
)

ex_adj_ms_step4_mean
```

```
##    param    Est     SE p_value   CI_lo CI_hi
##     beta 0.0776 0.0678 0.25251 -0.0553 0.211
##     tau2 0.0812 0.0317      NA  0.0378 0.175
##  lambda1 0.5189 0.1269 0.00731  0.3213 0.838
##  lambda2 0.3180 0.1214 0.00269  0.1505 0.672
```

Selection plot: Step 4PSM model mean only

```r
selection_plot(ex_adj_ms_step4_mean)
```



Moderator

Unadjusted estimate: moderator

```
ex_unadj_mod <- rma.uni(
  data = dat_ex,
  yi = d,
  sei = sd_d,
  method = "ML",
  mods = ~ stimuli
)
ex_unadj_mod
```

```
##
## Mixed-Effects Model (k = 96; tau^2 estimator: ML)
##
## tau^2 (estimated amount of residual heterogeneity):     0.0884 (SE = 0.0203)
## tau (square root of estimated tau^2 value):             0.2973
## I^2 (residual heterogeneity / unaccounted variability): 66.25%
## H^2 (unaccounted variability / sampling variability):   2.96
## R^2 (amount of heterogeneity accounted for):            9.01%
##
## Test for Residual Heterogeneity:
## QE(df = 94) = 274.4078, p-val < .0001
##
## Test of Moderators (coefficient 2):
## QM(df = 1) = 7.2714, p-val = 0.0070
##
## Model Results:
##
##               estimate      se     zval    pval    ci.lb    ci.ub
## intrcpt         0.3770  0.0706   5.3391  <.0001   0.2386   0.5155  ***
## stimuliwords   -0.2272  0.0843  -2.6966  0.0070  -0.3924  -0.0621   **
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ex_unadj_mod_cil <- ex_unadj_mod$tau2 - 1.96*ex_unadj_mod$se.tau2
ex_unadj_mod_ciu <- ex_unadj_mod$tau2 + 1.96*ex_unadj_mod$se.tau2
ex_unadj_mod_cil; ex_unadj_mod_ciu
```

```
## [1] 0.04870907
```

```
## [1] 0.1280912
```

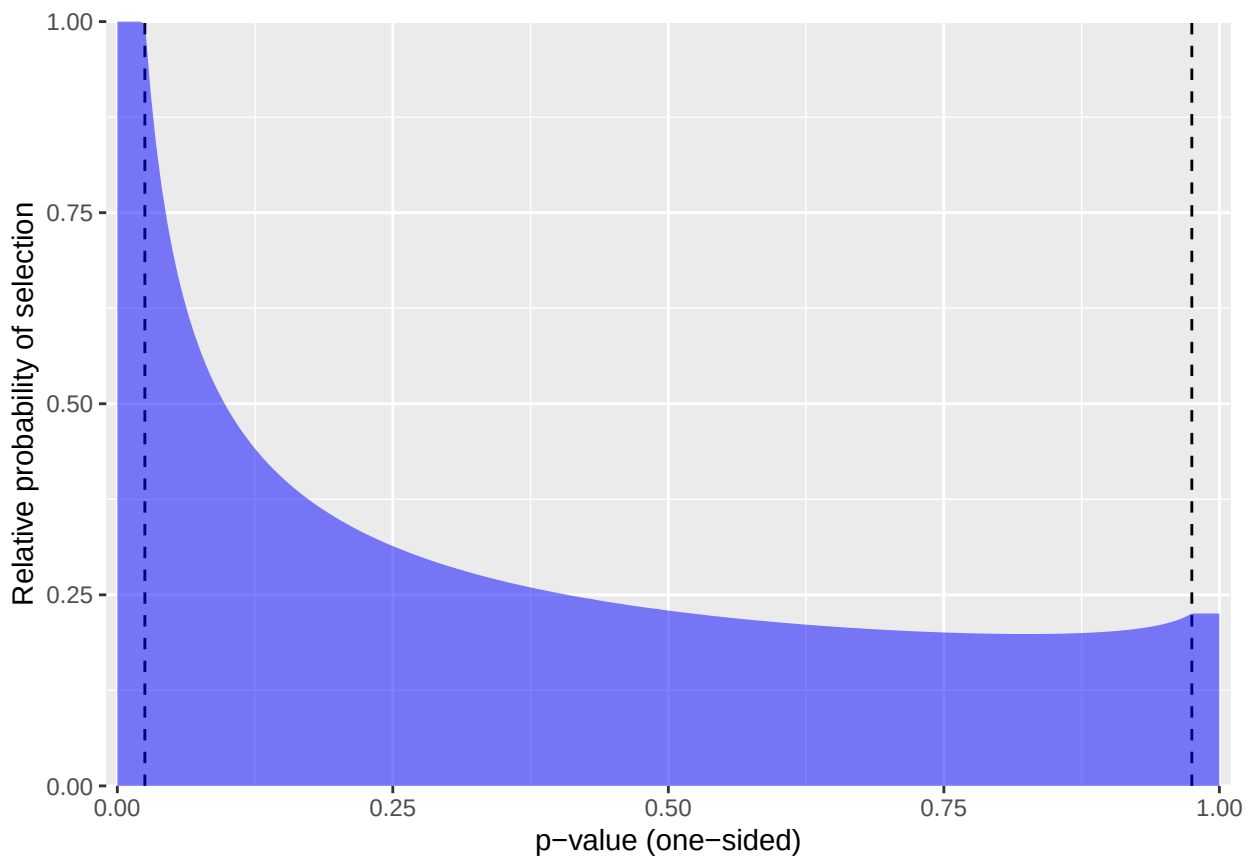
Adjusted estimate in metaselection: Beta model moderator

```r
ex_adj_ms_beta_mod <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  pi = p_onesided,
  cluster = studyid,
  selection_type = "beta",
  steps = c(.025,.975),
  make_sandwich = TRUE,
  mean_mods = ~ stimuli
)
ex_adj_ms_beta_mod
```

```
##              param     Est    SE p_value   CI_lo  CI_hi
##   beta_(Intercept)  0.1616 0.116  0.1630 -0.0654 0.3885
##  beta_stimuliwords -0.1937 0.120  0.1078 -0.4297 0.0424
##               tau2  0.0639 0.021      NA  0.0335 0.1218
##            lambda1  0.4842 0.173  0.0429  0.2399 0.9772
##            lambda2  0.8906 0.199  0.6049  0.5743 1.3812
```

Selection plot: Beta model moderator

```r
selection_plot(ex_adj_ms_beta_mod)
```



Adjusted estimate in metaselection: Step 3PSM model moderator

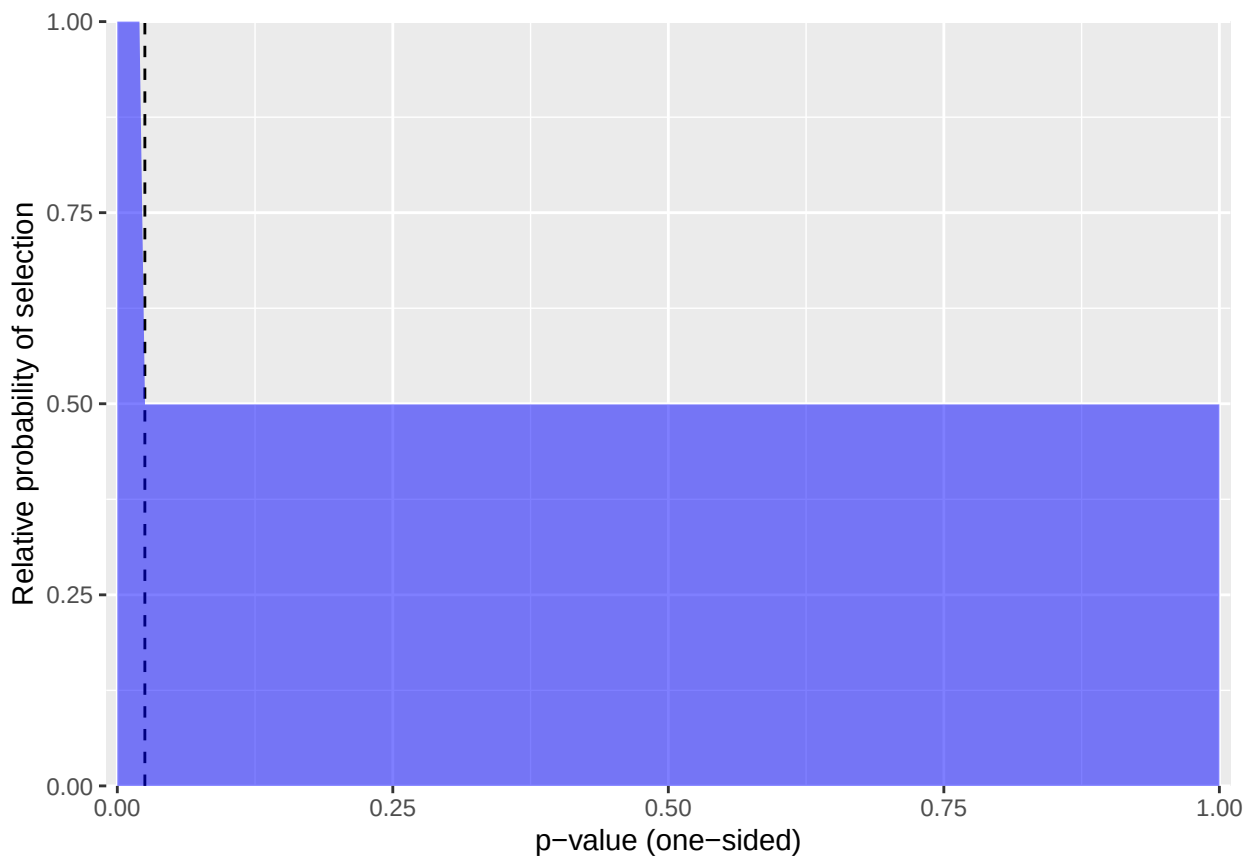
```
ex_adj_ms_step3_mod <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = 0.025,
  mean_mods = ~ stimuli
)

ex_adj_ms_step3_mod
```

```
##              param     Est     SE p_value   CI_lo  CI_hi
##   beta_(Intercept)  0.2747 0.0908 0.00248  0.0967 0.4526
##  beta_stimuliwords -0.1961 0.1059 0.06419 -0.4038 0.0116
##               tau2  0.0657 0.0244      NA  0.0318 0.1359
##            lambda1  0.4992 0.1408 0.01376  0.2872 0.8676
```

Selection plot: Step 3 PSM model moderator

```
selection_plot(ex_adj_ms_step3_mod)
```



Adjusted estimate in metaselection: Step 4PSM model moderator

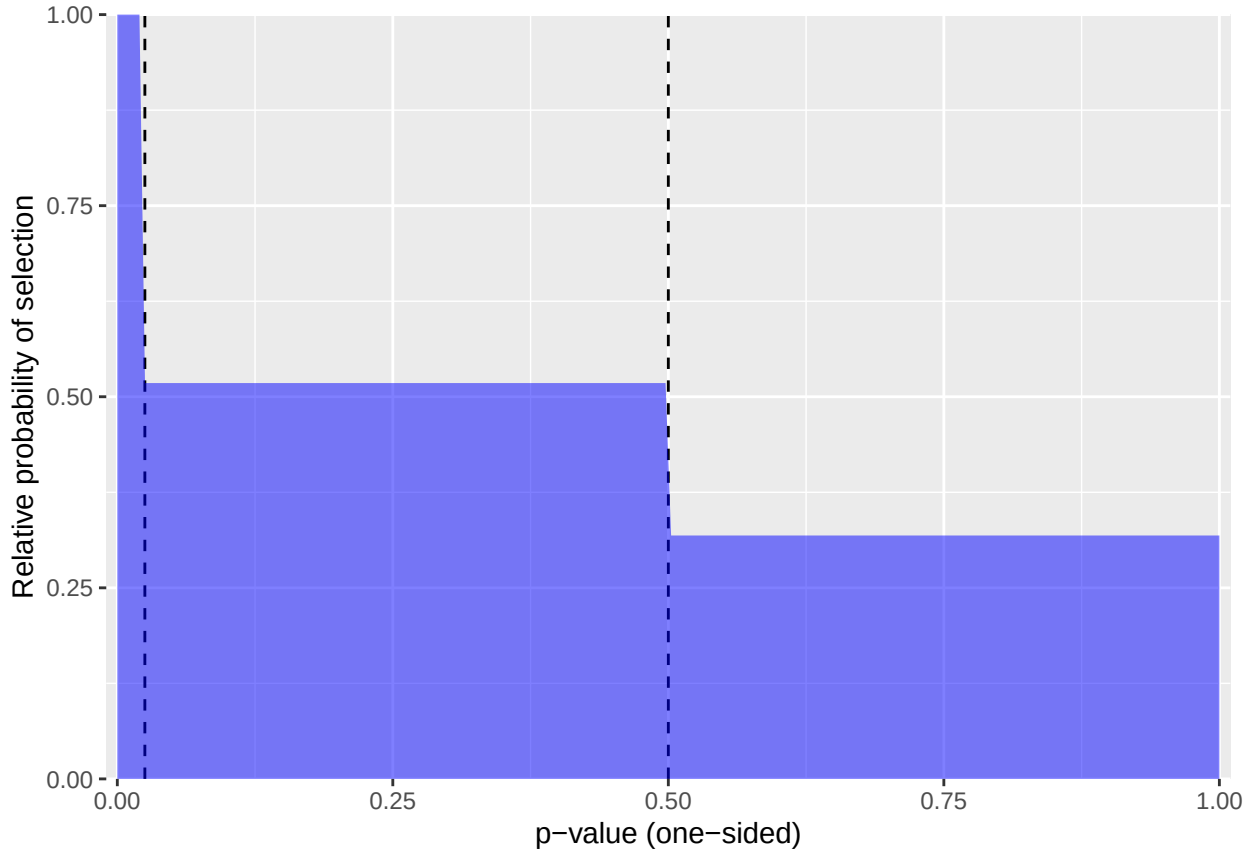
```
ex_adj_ms_step4_mod <- selection_model(
  dat = dat_ex,
  yi = d,
  sei = sd_d,
  cluster = studyid,
  selection_type = "step",
  steps = c(0.025, .5),
  mean_mods = ~ stimuli
)

ex_adj_ms_step4_mod
```

```
##              param     Est     SE p_value   CI_lo  CI_hi
##   beta_(Intercept)  0.2321 0.0997 0.01992  0.0367 0.4275
##  beta_stimuliwords -0.2119 0.1167 0.06936 -0.4405 0.0168
##               tau2  0.0744 0.0282      NA  0.0354 0.1562
##            lambda1  0.5179 0.1232 0.00569  0.3249 0.8257
##            lambda2  0.3185 0.1204 0.00248  0.1518 0.6683
```

Selection plot: Step 4PSM model moderator

```
selection_plot(ex_adj_ms_step4_mod)
```


1.00
0.75
0.50
0.25
0.00
Relative probability of selection
0.00
0.25
0.50
0.75
1.00
p−value (one−sided)